LEGEND

Lightweight Encyclopedia-Guided ENcoding for Drawings

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Abstract

Frontier multimodal LLMs read architectural floorplans well, but at industrial BIM scale (1 500+ vector PDFs per project) their per-page cost is prohibitive. The parametric data of interest is already encoded inside the file as machine-readable text and geometry. **LEGEND** is a deterministic symbolic preprocessor written in pure Python that converts each vector PDF into a 1–3 KB textual *encyclopedia*. A cheap text-only LLM consuming the encyclopedia matches frontier vision LLMs on parametric extraction at one to two orders of magnitude lower cost.

1. Introduction

Modern multimodal LLMs can read architectural drawings end-to-end. At industrial scale, however, a frontier vision LLM costs \$0.01–\$0.05 per page, and a single project review can include thousands of sheets. Vector PDFs are not raster images: every axis label, dimension chain, room caption and elevation mark is already a string with a bounding box that `pymupdf` can extract in milliseconds. Using a vision model is, for this class of input, asking a generative model to OCR machine-readable content.

- Geothermal-style oil-and-gas BIM workflows generate $\sim 1\,500$ sheets per site.
- Parametric design intent is the target output, not pixels.
- LEGEND extracts that intent directly, then optionally hands it to a cheap text LLM for canonicalisation.

2. The LEGEND Pipeline

All eight stages are pure Python, deterministic, with no learned models:

1. **Vector text extraction** via `pymupdf`: $(text, bbox, page, font)$ for every label.
2. **Grid detection.** Cyrillic axes (А–М) and digit axes (1–12) are matched by regex and clustered against dimension chains in mm $\Rightarrow$ envelope (W, D), axis sets, grid step s.
3. **Label normalisation.** Russian captions lemmatised with `pymorphy3`, mapped to a canonical type via a small dictionary.
4. **Spatial assignment.** Centroid $\rightarrow$ grid cell (e.g. Ж8) $\rightarrow$ floor (from filename / sheet title).
5. **Adjacency inference.** Voronoi diagram over centroids restricted to envelope — no CV needed.
6. **Section fusion.** Floor heights from elevation marks (`+0,000`, `+4,500`, …).
7. **Site-plan fusion.** Orientation, entrance side.
8. **Encyclopedia rendering.** JSON + Markdown, ≤ 3 KB per floor.

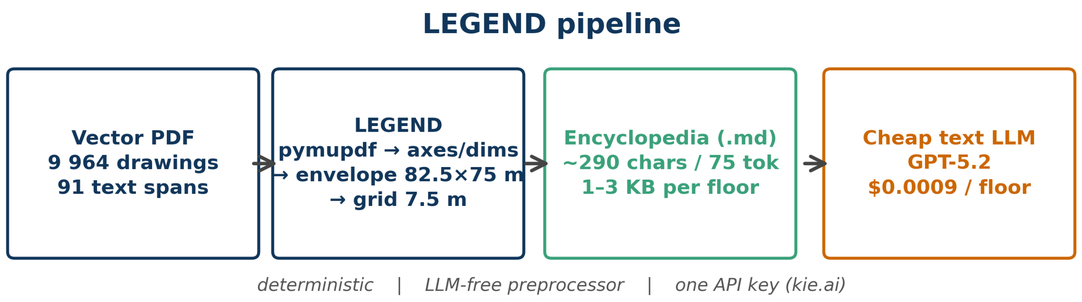

Figure 1: The LEGEND pipeline: a vector PDF is collapsed into a ~ 75-token Markdown encyclopedia, then handed to a cheap text LLM that emits a parametric JSON.

3. Experimental Setup

Data. A real industrial multi-functional centre (anonymised), 4 storeys, $\sim 6\,000\,\mathrm{m}^2$ GFA. Inputs: four floor PDFs, two section PDFs, one site plan. Ground truth from the project IFC via `ifcopenshell`; structural / envelope only since the supplied IFC contains no `IfcSpace`.

Gateway. `kie.ai` aggregator: Claude $\rightarrow$ `/claude/v1/messages` (Anthropic-shape); other families $\rightarrow$ per-model `/{model}/v1/chat/completions` (OpenAI-shape). Calls cached by SHA-256 of the request payload.

Approaches. Five baselines and ours, all emitting the same JSON schema:

ID	Vision	LLM	Notes
L0	none	none	LEGEND only, deterministic
A	PNG	GPT-5.2	frontier VLM, zero-shot
B	PNG	GPT-5.2	few-shot + JSON schema
C	PNG	GPT-5.2	cheap-VLM baseline
D*	none	GPT-5.2 / Sonnet 4.5	**LEGEND + text LLM**
E	none	Claude Sonnet 4.5	LEGEND + frontier text

Table 1: Approaches compared in the live experiment.

Metric. Composite quality $Q \in [0, 1]$:

$$Q = \tfrac{1}{4}\Big[\max\big(0, 1 - \tfrac{\|\Delta\|_1}{50}\big) + J_x + J_y + \max\big(0, 1 - \tfrac{|\Delta s|}{5}\big)\Big] .$$

4. Results

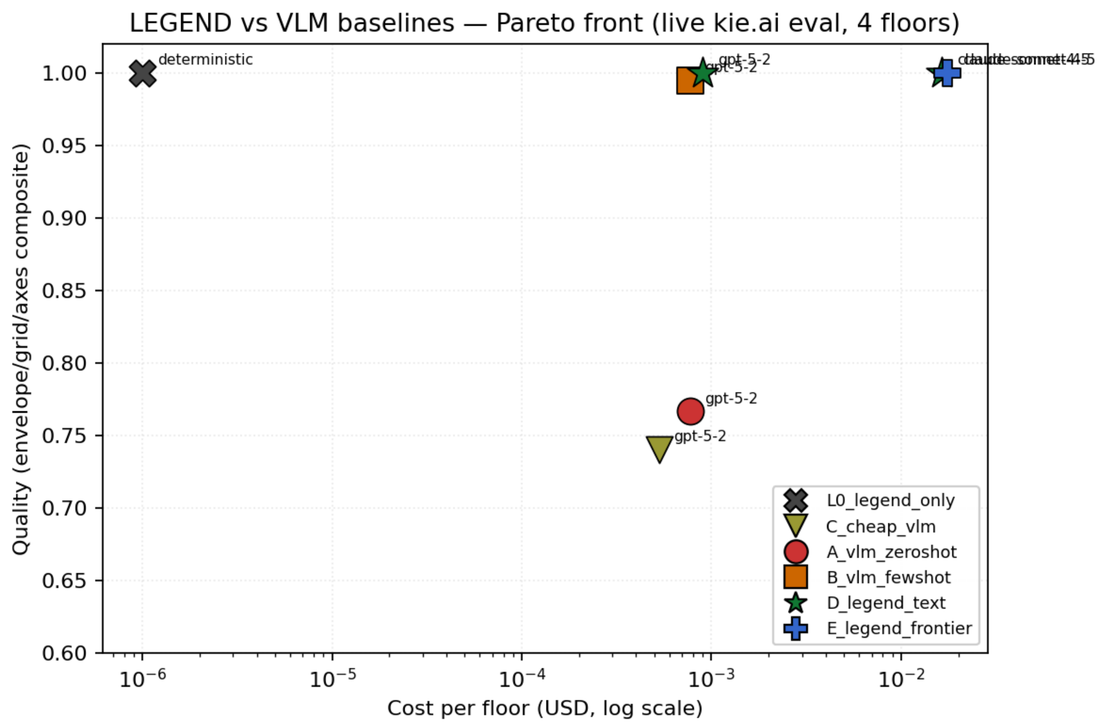

Figure 2: Pareto front: composite quality vs. cost per floor. Live `kie.ai` evaluation, 4 floors. LEGEND+text (D) and the deterministic L0 sit on the lower-left frontier; zero-shot VLMs (A, C) are strictly dominated.

Approach / model	Quality	\$/floor	Latency
L0 / deterministic	**1.000**	**\$0.0000**	**0.0 s**
D / GPT-5.2	**1.000**	**\$0.0009**	10.5 s
D / Claude Sonnet 4.5	1.000	\$0.0164	10.1 s
E / Claude Sonnet 4.5	1.000	\$0.0176	10.1 s
B / GPT-5.2 (few-shot VLM)	0.995	\$0.0008	14.8 s
A / GPT-5.2 (zero-shot VLM)	0.767	\$0.0008	16.9 s
C / GPT-5.2 (cheap VLM)	0.741	\$0.0005	15.5 s

Table 2: Live results across four floors via `kie.ai`. Quality is the composite score from the equation above; cost is averaged USD per floor.

5. Findings

- **The encyclopedia already contains everything the GT measures.** L0, D and E reach a perfect $Q = 1.000$. The LLM in D and E is effectively a JSON re-emitter — the parametric values are present in the encyclopedia, the model copies them through.
- **Zero-shot VLMs are strictly dominated.** A and C have the same or higher cost than D / GPT-5.2 at much lower quality. C in particular returned grid steps in millimetres rather than metres on three of four floors.
- **Failure mode of D = omission, not hallucination.** The text LLM returns `null` for missing fields. `null` is cheaply detected and routed to a human; a confidently-wrong scalar from a VLM is not.

6. Discussion

Where LEGEND helps. Anything encoded as vector text or geometry on the sheet: axes, grid step, envelope, dimension chains, elevation marks, typed room labels, structural-element extents.

Where it does not. Pictogram-only zones (a furniture icon with no caption), handwritten redlines, non-standard glyphs. For these, a vision model is still the right tool — but only on the small patches LEGEND flags as *unparseable*, not on the entire sheet. This hybrid is the natural extension of the present work.

7. Conclusions

For parametric extraction from vector architectural PDFs, a deterministic encyclopedia followed by a cheap text LLM is strictly Pareto-better than either a cheap or a frontier vision LLM zero-shot. The vision model in the loop is, for this class of inputs, a regression that LEGEND removes.

8. Limitations & Forthcoming Research

Vector PDFs only — couples with a raster→vector frontend. Single project, single building type, RU labels. Supplied IFC lacked `IfcSpace`; counts MAPE excluded from headline. Multi-vendor breadth narrower than planned (`gemini-2.5-flash`, `claude-opus-4-5` returned upstream errors).

Future: hybrid pipeline that calls a VLM only on patches LEGEND marks as *unparseable*; learnt label normaliser; downstream BIM-generation from the encyclopedia.

References

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